Analytic Rings

Nir Elber

Fall 2026

Contents

Contents **1**

1 September 21: Analytic Stacks **1**
1.1 Condensed Math 1
1.2 Analytic Rings 3

2 September 28: The Four Functors **4**
2.1 Six Functors on Analytic Rings 4
2.2 The Analytic Ring $\mathbb{Z}[T]_{\blacksquare}$ 6

3 October 19: Gelfand Rings **8**
3.1 Overconvergent Elements 8
3.2 Uniform Rings 9
3.3 Gelfand Rings 10
3.4 The Berkovich Spectrum 11

4 October 26: Gelfand Rings and Gelfand Stacks **11**
4.1 Fredholm Rings 11
4.2 More on the Berkovich Spectrum 12
4.3 Gelfand Stacks 13

1 September 21: Analytic Stacks

This talk was given by David Yang at MIT for the analytic stacks seminar.

1.1 Condensed Math

Let's start with a review of condensed mathematics. We would like to do topology and homological algebra simultaneously, but this is not possible classically. For example, the category of topological abelian groups does not form an abelian category: one can embed $\mathbb{R}$ with the discrete topology into $\mathbb{R}$ with the usual topology, and there is no reasonable cokernel for this map.

The solution to this classical problem proposed by condensed mathematics is to embed topological abelian groups into a larger category.

Definition 1 (light)**.** A profinite set is *light* if and only if it is separable.

Remark 2. Due to compactness, separability is equivalent to being first countable. It also happens to be equivalent to metrizability, and it is equivalent to being an inverse limit of finite sets indexed by $\mathbb{N}$.

Definition 3. A *covering* of a light profinite set is a finite collection of jointly surjective light profinite sets. A light *condensed set* is a sheaf on the category of light condensed sets. We let Cond be the category of condensed sets, and we let AbCond be the category of abelian condensed sets.

Remark 4. To give a condensed set is to give a presheaf $\mathcal{F}$ on light profinite sets for which $\mathcal{F}(S \sqcup T) = \mathcal{F}(S) \times \mathcal{F}(T)$, and any surjective covering $S \twoheadrightarrow T$ induces

$$\mathcal{F}(T) = \mathrm{eq}(\mathcal{F}(S) \to \mathcal{F}(S \times_T S)).$$

Example 5. For any topological space X, there is a condensed set $\mathrm{Mor}_{\mathrm{Top}}(-, X)$. Similarly, an abelian topological group defines an abelian condensed set. The functor from topological spaces to condensed sets is fully faithful on a large subcategory (e.g., metrizable, compactly generated, Hausdorff spaces).

We now pass to solid condensed sets. The adjective "solid" is intended to force us to work with nonarchimedean geometry. For motivation, note that a sequence $\{x_i\}_i$ in $\mathbb{R}$ (or any metric abelian group) which is summable is automatically a null sequence, but the converse need not hold. However, the converse does hold over $\mathbb{Q}_p$. Thus, we will want to say something about convergence.

Notation 6. Fix a condensed set S. Then define $\mathbb{Z}[S]$ as the sheafification of the presheaf

$$T \mapsto \mathbb{Z}[S(T)].$$

Notation 7. We fix $P \coloneqq \mathbb{Z}[\mathbb{N} \cup \{\infty\}]/\mathbb{Z}[\{\infty\}]$, where $\mathbb{N} \cup \{\infty\}$ is the one-point compactification of $\mathbb{N}$. The continuous map $(+1)\colon \mathbb{N} \cup \{\infty\} \to \mathbb{N} \cup \{\infty\}$ induces an automorphism $[1]\colon P \to P$.

The moral is that P represents null sequences.

Lemma 8. Fix a topological abelian group A. Then $\mathrm{Hom}(P, A)$ is equivalent to the group of null sequences in A.

Proof. Look at where each point goes. By construction, P sends ∞ to $0 \in A$. ∎

Lemma 9. The condensed abelian group P is projective.

Proof. Expand out the definitions. ∎

Definition 10 (solid)**.** A condensed abelian group A is *solid* if and only if the map

$$(\mathrm{id} - 1)\colon \mathrm{Hom}(P, A) \to \mathrm{Hom}(P, A)$$

is an isomorphism.

Remark 11. Intuitively, $\mathrm{id} - 1$ on null sequences is given by

$$(a_0, a_1, \ldots) \mapsto (a_0 - a_1, a_1 - a_2, \ldots).$$

This is inverted by some telescoping map, which is where summability enters the picture.

Example 12. It turns out that $\mathbb{Z}$ is solid. The idea is that a null sequence in $\mathbb{Z}$ eventually vanishes and is thus summable. This shows that solid condensed abelian groups does not form a $\otimes$-ideal.

Remark 13. The full subcategory of solid condensed abelian groups is stable under limits, colimits, and extensions.

Example 14. The forgetful functor from solid condensed abelian groups to abelian groups admits a right adjoint. For example, the solidification of $\mathbb{R}$ vanishes.

Remark 15. The functor from the derived category of solid condensed abelian groups to condensed abelian groups is fully faithful, and the image is given by those complexes with solid cohomology groups. For example, if C is solid, then for any condensed abelian group A, the complex $\mathrm{RHom}(A, C)$ is solid.

1.2 Analytic Rings

Here is our definition.

Definition 16 (analytic ring)**.** An *analytic ring* is a pair $(A^{\triangle}, D(A))$, where $A^{\triangleleft}$ is a condensed ring and $D(A) \subseteq D(A^{\triangleleft})$, for which

- $A^{\triangleleft} \in D(A)$,
- $D(A)$ is stable under limits and colimits (and so $D(A) \subseteq D(A^{\triangleleft})$ admits a left adjoint F),
- for any $C \in D(\mathrm{CondAb})$ and $M \in D(A)$, one has $\mathrm{RHom}_{\mathrm{CondAb}}(C, M) \in D(A)$, and
- F sends connected objects to connected objects (namely, $D(A^{\triangleleft})^{\leq 0}$ goes to $D(A^{\triangleleft})$).

Without the first condition, we may refer to $(A^{\triangleleft}, D(A))$ as an *uncompleted analytic ring*.

Example 17. Our discussion of solidity shows that $(\mathbb{Z}, D(\mathrm{SolCondAb}))$ is an analytic ring.

Example 18. There is an analytic ring with $A^{\triangleleft} = \mathbb{R}$ and $D(A)$ given by the "liquid" condensed abelian groups.

Example 19. For any condensed ring A, one can take $D(A) \coloneqq D(A^{\triangleleft})$.

Example 20. There is an analytic ring attached to any Huber ring (A, A^{+}).

Remark 21. The second condition implies that $D(A)$ is symmetric monoidal, and the functor F is symmetric monoidal. The idea is to show that $\ker F$ is a $\otimes$-ideal. Well, for $B \in \ker F$ and $C \in D(A^{\triangleleft})$, we must show that $F(B \otimes C) = 0$. This is equivalent to checking $\mathrm{Hom}_{D(A)}(F(B \otimes C), E) = 0$ for any $E \in D(A)$, but

$$\mathrm{Hom}_{D(A)}(F(B \otimes C), E) = \mathrm{Hom}_{D(A^{\triangleleft})}(B \otimes C, E) = \mathrm{Hom}_{D(A^{\triangleleft})}(B, \mathrm{Hom}_{D(A^{\triangleleft})}(C, E)).$$

We will thus be done by the free–forgetful adjunction if we can show that $\mathrm{Hom}_{D(A^{\triangleleft})}(C, E)$ lives in $D(A)$. Well, it suffices to show this where C is a set of generators of $D(A^{\triangleleft})$, for which we can take to be objects of the form $A^{\triangleleft} \otimes_{\mathbb{Z}} S$, and now we are done by the second condition.

The last thing we will talk about today is colimits.

Theorem 22. Colmits exist in the category of analytic rings.

Proof. First, one shows that colimits exist in the category of uncompleted analytic rings. The underlying condensed ring is given by taking the underlying colimits, and the underlying derived catgegory is the subcategory of objects which land in the derived category of each component.

Second, one shows that the forgetful functor from analytic rings to uncompleted analytic rings admits a left adjoint, so the right adjoint will give us our colimits. We will merely give the construction: for an uncompleted analytic ring $(A^\blacktriangleleft, D(A))$, simply replace $A^\blacktriangleleft$ to its image along the free functor $D(A^\blacktriangleleft) \to D(A)$. ■

Example 23. Fix an analytic ring $(A^\blacktriangleleft, D(A))$, and let $B^\blacktriangleleft$ be a condensed $A^\blacktriangleleft$-algebra such that $B^\blacktriangleleft \in D(A)$. Then $(B^\blacktriangleleft, \mathrm{Mod}_{B^\blacktriangleleft} D(A))$ is the colimit of the following diagram.

$$\begin{array}{ccc} (A^\blacktriangleleft, D(A)) & \longrightarrow & (B^\blacktriangleleft, D(B^\blacktriangleleft)) \\ \downarrow & & \downarrow \\ (A^\blacktriangleleft, D(A)) & \longrightarrow & (B^\blacktriangleleft, \mathrm{Mod}_{B^\blacktriangleleft} D(A)) \end{array}$$

Here, one can directly define $\mathrm{Mod}_{B^\blacktriangleleft} D(A)$ as the subcategory of $D(B^\blacktriangleleft)$ which is the pre-image of $D(A) \subseteq D(A^\blacktriangleleft)$.

2 September 28: The Four Functors

This talk took place at MIT for the Rigid analytic rings seminar.

2.1 Six Functors on Analytic Rings

The first part of the talk was given by David.

Last time, we defined an analytic ring as a pair $(A^\blacktriangleleft, D(A))$ satisfying some properties. We start by remarking that $D(A)$ is more important than $A^\blacktriangleleft$.

Example 24 (categorical Künneth)**.** Fix a diagram $A_\bullet$ from an index category $\mathcal{I}$ to the category of analytic rings, and let $(A^\blacktriangleleft, D(A))$ be its colimit. Last time, we checked that

$$D(A) \cong \operatorname*{colim}_i D(A_i),$$

where the colimit is taken in the category of presentable categories. However, the underlying $A^\blacktriangleleft$ had to be modified rather significantly from $\mathrm{colim}_i A_i^\blacktriangleleft$.

Remark 25. Colimits can be turned into limits. In short, by the Adjoint functor theorem, the internal morphisms $(A^\blacktriangleleft, D(A)) \to (B^\blacktriangleleft, D(B))$ have $D(B) \to D(A)$ with a left adjoint $D(A) \to D(B)$. (In short, the adjoint is given by completing $(- \otimes_{A^\blacktriangleleft} B^\blacktriangleleft)$.) Then reversing all arrows turns colimits into limits. For example, one finds that

$$D(A) = \lim_i D(A_i)$$

after enough abuse of notation.

Our present goal is to produce a six-functor formalism on analytic rings. In light of our intuition that $D(A)$ is more important than $A^\blacktriangleleft$, this will mostly be done categorically.

Definition 26. For a morphism $f\colon \operatorname{Sp} B \to \operatorname{Sp} A$ of analytic rings, we define the $*$-pushforward as the underlying map $f_*\colon D(B) \to D(A)$, and its right adjoint is the $*$-pullback f^*.

For the remaining functors, we need classes of maps I, P, and E. In short, I should consist of open embeddings, P should consist of proper maps, and E should be !-able maps. Notably, following Nagata compactification, E is the same as composites of a map in I with a map in P.

Let's start with P.

Definition 27. A map $f\colon A \to B$ of analytic rings is *proper* if and only if

$$\begin{array}{ccc} D(B) & \longrightarrow & D(B^{\blacktriangleleft}) \\ \downarrow & & \downarrow \\ D(A) & \longrightarrow & D(A^{\blacktriangleleft}) \end{array}$$

is a pullback square, where the horizontal arrows are the completion functors.

Remark 28. It is equivalent to require that $D(B) \cong \operatorname{Mod}_{B^{\blacktriangleleft}} D(A)$.

Remark 29. If $f\colon A \to B$ is proper, then the $*$-pullback map $D(A) \to D(B)$ is just $- \otimes_{A^{\blacktriangleleft}} B^{\blacktriangleleft}$.

Definition 30. Fix a proper map $f\colon A \to B$ of analytic rings. Then the right adjoint of $f^*\colon D(A) \to D(B)$ is denoted $f_!\colon D(B) \to D(A)$.

Remark 31. One can check that base-change and the projection formula both hold for $f_!$ in this setting.

We now turn to I. Here is the analytic example.

Example 32. Let $j\colon X \to Y$ be an open embedding. Then the functor j^* on constructible sheaves admits a fully faithful left adjoint, and it satisfies the projection formula

$$j_!(j^* A \otimes B) = A \otimes j_! B.$$

Remark 33. If one works with quasicoherent sheaves instead of constructible sheaves, then everything appears backwards.

Remark 34. There is a similar calculation that can be done for closed embeddings, which have fully faithful right adjoints.

Let's turn this into a definition.

Definition 35. A map $f\colon \mathcal{C} \to \mathcal{D}$ of stable categories is *open* if and only if it admits a fully faithful left adjoint and satisfying some projection formula.

Example 36. Let A be an idempotent algebra in $\mathcal{C}$, meaning that $A \otimes A \cong A$. Then the forgetful functor $\operatorname{Mod}_A(\mathcal{C}) \to \mathcal{C}$ is fully faithful and admits the adjoint given by the tensor product. It turns out that the adjoint construction provides all the closed embeddings.

A similar example works for open embeddings.

Definition 37. A map $j\colon A \to B$ of analytic rings is *open* if and only if $j^*\colon D(A) \to D(B)$ is open. By definition, we are provided with a left adjoint $j_!\colon D(B) \to D(A)$.

At long last, we have enough functors to state a theorem.

Theorem 38. The functors $(\otimes, f_*, f_!)$ defines a three-functor formalism satisfying the projection formula and base-change.

This is not totally formal. For example, one needs some base-change property for proper and open maps. However, we will not explicate any of these checks.

We close our discussion by defining an analytic stack.

Definition 39. The !-topology on the opposite category of analytic rings contains the covers $\{B \to A_i\}_i$ of !-able maps where $D(B)$ is isomorphic to the !-limit of the simplicial object

$$D(B) \cong \lim D^!(A_i^{\coprod_B n}),$$

where the right-hand side is the simplicial object. Here, we are taking !-pullbacks, which is the right adjoint of !-pushforward.

Remark 40. One can check that the !-topology is finitary and subcanonical.

Definition 41. A map $X \to Y$ in $\mathrm{PSh}(\mathrm{AnRing}_!^{\mathrm{op}})$ is a !-equivalence if and only if any map $M \to N$ in $\mathrm{PSh}(\mathrm{AnRing}_!^{\mathrm{op}})$ which is a pullback of an iterated diagonal of $X \to Y$, we have that

$$D(N) \cong \lim D^!(M^{\times_N n}),$$

where the right-hand side is the simplicial object.

Example 42. If X and Y are already analytic rings, then being a !-equivalence is equivalent to $X \to Y$ being

Definition 43 (analytic stack)**.** A functor $\mathcal{F}\colon \mathrm{AnRing}^{\mathrm{op}} \to \mathrm{Sp}$ is an analytic stack if and only if any !-equivalence $X \to Y$ makes the map $\mathrm{Hom}(Y, \mathcal{F}) \to \mathrm{Hom}(X, \mathcal{F})$ into an isomorphism.

2.2 The Analytic Ring $\mathbb{Z}[T]_\blacksquare$

This second part of the talk was given by Tong. The goal of the second part of the talk is to get some intuition about the analytic ring $\mathbb{Z}[T]_\blacksquare$. Recall the following definition.

Definition 44. A condensed abelian group A is *solid* if and only if the map $\mathrm{id} - 1$ is an isomorphism on $\mathrm{Hom}_{\mathbb{Z}}(P, M)$, where $P = \mathbb{Z}[\mathbb{N} \cup \{\infty\}]/\mathbb{Z}[\{\infty\}]$.

We now introduce $\mathbb{Z}_\blacksquare[T]$, which is almost $\mathbb{Z}[T]_\blacksquare$.

Notation 45. Define the analytic ring $\mathbb{Z}_\blacksquare[T]$ to consist of the ring $\mathbb{Z}[T]$ along with the subcategory $D(\mathbb{Z}_\blacksquare[T]) \subseteq D(\mathbb{Z}[T])$ consisting of the solid modules.

Approximately speaking, $\mathbb{Z}[T]_\blacksquare$ should be like an affine line over $\mathbb{Z}_\blacksquare$. We would like to construct a unit disk over $\mathbb{Z}_\blacksquare$. In classical rigid analytic geometry, this is done over $\mathbb{Q}_p$ via the Tate algebra $\mathbb{Q}_p\langle T\rangle$ of series $\sum_n a_n T^n$ whose coefficients tend to zero. Thus, we expect that

$$\mathbb{Z}[T]_\blacksquare \otimes_{\mathbb{Z}_\blacksquare} \mathbb{Q}_p \cong \mathbb{Q}_p\langle T\rangle.$$

For example, one would expect that $\mathbb{Z}[T]_\blacksquare$ still has underlying ring $\mathbb{Z}[T]$, and we will just change the underlying category. Also taking motivation from our remark about $\mathbb{Q}_p\langle T\rangle$, we have the following definition.

Definition 46. Define the pair $\mathbb{Z}[T]_\blacksquare$ to have underlying ring $\mathbb{Z}[T]$ and category $D(\mathbb{Z}[T]_\blacksquare)$ consisting of those solid $M \in D(\mathbb{Z}[T])$ for which any nilsequence $\{a_n\}$ in M makes $\sum_n a_n T^n$ convergent. This latter condition means that

$$(\mathrm{id} - T)\colon \mathrm{Hom}_{\mathbb{Z}_\blacksquare[T]}(P', M) \to \mathrm{Hom}_{\mathbb{Z}_\blacksquare[T]}(P', M)$$

is an isomorphism, where $P' = P \otimes_{\mathbb{Z}} \mathbb{Z}_\blacksquare[T]$.

Remark 47. One can check that P' is isomorphic to $\mathbb{Z}[[q]][T]$ (equipped with the q-adic topology). Using the exactness of the functor from locally compact abelian groups to condensed abelian groups, one finds that the sequence

$$0 \to \mathbb{Z}[[q]][T] \xrightarrow{1-qT} \mathbb{Z}[[q]][T] \to \mathbb{Z}((1/T)) \to 0$$

is exact. Thus, $M \in D(\mathbb{Z}_\blacksquare[T])$ lives in $D(\mathbb{Z}[T]_\blacksquare)$ if and only if $\mathrm{Hom}_{\mathbb{Z}_\blacksquare[T]}(\mathbb{Z}((1/T)), M) = 0$.

Remark 48. One can compute that $\mathbb{Z}((1/T))$ is an idempotent $\mathbb{Z}_\blacksquare[T]$-algebra. Thus, there is an idempotent decomposition

$$D(\mathbb{Z}((1/T))) \to D(\mathbb{Z}_\blacksquare[T]) \to D(\mathbb{Z}[T]_\blacksquare),$$

where the former map is given by $(-)_*$. For example,

Let's now check that $\mathbb{Z}[T]_\blacksquare$ is an analytic ring.

Theorem 49. The pair $\mathbb{Z}[T]_\blacksquare$ is an analytic ring. Also, for any light profinite set $S = \lim S_n$, one has

$$\mathbb{Z}[T]_\blacksquare[S] = \lim_n \mathbb{Z}[T][S_n].$$

Proof. We only discuss the second statement. It turns out to reduce to $\prod_{\mathbb{N}} \mathbb{Z}$, and the key calculation is that the cokernel of the map

$$\left(\prod_{\mathbb{N}} \mathbb{Z}\right)[T] \subseteq \prod_{\mathbb{N}} (\mathbb{Z}[T])$$

is a $\mathbb{Z}((1/T))$-module. In fact, it is

$$\left(\prod_{\mathbb{N}} \mathbb{Z}((1/T))\right) \Big/ \left(\prod_{\mathbb{N}} (\mathbb{Z} \otimes_{\mathbb{Z}_\blacksquare} \mathbb{Z}((1/T))\right).$$

(Here, recall $\otimes_{\mathbb{Z}_\blacksquare}$ is the solidification of the usual tensor product.) One sees this by doing some combinatorics on the support of these power series. ∎

Example 50. To close the talk, we sketch that the $\mathbb{Z}[T]_\blacksquare$-completion of $\mathbb{Q}_p \otimes_{\mathbb{Z}_\blacksquare} \mathbb{Z}[T]_\blacksquare$ is $\mathbb{Q}_p\langle T\rangle$ as $\mathbb{Z}_\blacksquare$-modules. Well, $\mathbb{Z}_p \otimes_{\mathbb{Z}_\blacksquare} \mathbb{Z}[T]_\blacksquare$ can be computed as $\mathbb{Z}_p = \lim_n \mathbb{Z}/p^n\mathbb{Z}$. It turns out that $\otimes_{\mathbb{Z}_\blacksquare}\mathbb{Z}[T]_\blacksquare$ commutes with limits, so

$$\mathbb{Z}_p \otimes_{\mathbb{Z}_\blacksquare} \mathbb{Z}[T]_\blacksquare \cong \lim_n (\mathbb{Z}/p^n\mathbb{Z}) \otimes_{\mathbb{Z}_\blacksquare} \mathbb{Z}[T]_\blacksquare \cong \lim \mathbb{Z}[T]/p^n\mathbb{Z}[T],$$

where the last isomorphism requires no solidification because $\mathbb{Z}[T]/p^n\mathbb{Z}[T]$ is already solid (indeed, it is discrete). This last limit is just $\mathbb{Z}_p\langle T\rangle$. To compute $\mathbb{Q}_p \otimes_{\mathbb{Z}_\blacksquare} \mathbb{Z}[T]$, we now take the colimit as $\mathbb{Q}_p = \operatorname{colim} p^n\mathbb{Z}_p$ and pass the colimit out of the tensor product.

3 October 19: Gelfand Rings

This talk was given by Daniel at MIT for the Analytic rings seminar.

3.1 Overconvergent Elements

Last time, we defined the idempotent algebra $\mathbb{Q}_p\langle \mathbb{N}[S]\rangle_{\leq r}$.

Definition 51. For each $A \in \mathrm{Ring}_{\mathbb{Q}_p}$, define the condensed subring $A^{\leq r} \subseteq A$ for the subring given by sending profinite sets T to

$$\mathrm{Mor}_{\mathrm{Ring}_{\mathbb{Q}_p}}(\mathbb{Q}_p\langle \mathbb{N}[S]\rangle_{\leq r}, A) \subseteq A(T).$$

We define the *nilradical* to be $\mathrm{Nil}^\dagger A \coloneqq A^{\leq 0}$. While we're here, we define $A^{<r} \coloneqq \bigcup_{r'<r} A^{\leq r'}$.

Remark 52. It turns out that $A^{\leq r}$ as defined is a sheaf on ProFin. (The prefix "hyper" is required because everything is derived.)

Remark 53. Given a ring R, one can define $R^\times$ as $\mathrm{Hom}(\mathbb{Z}[T,1/T],R)$. Thus, we may think of the ring $\mathbb{Z}[T,1/T]$ as testing units. Analogously, $\mathbb{Q}_p\langle \mathbb{N}[S]\rangle_{\leq r}$ tests for size bounded by r.

Lemma 54. Fix $A \in \mathrm{Ring}_{\mathbb{Q}_p}$. For any $r \geq 0$, the condensed ring $A^{\leq r}$ is solid.

Proof. Last time, we realized that $\mathbb{Q}\langle \mathbb{N}[S]\rangle$ as a Tate algebra with enough variables after extending $\mathbb{Q}$. We can do a similar trick this time, but to keep track of the power-bounding, we need to do an integral variant. Namely, one has

$$\mathbb{Z}_p\left[p^{\mathbb{Q}^+}\right] \otimes_{\mathbb{Z}_p} A^{\leq r} = \bigcap_{\substack{a\in\mathbb{Q}\\ p^{-a}\leq r}} p^{-a}\left(\mathbb{Z}_p\left[p^{\mathbb{Q}^+}\right] \otimes_{\mathbb{Z}_p} A^\circ\right).$$

Now, A° is solid, so each disk is solid, so the entire intersection is solid, so the retract $A^{\leq r}$ is solid. ∎

Example 55. Let A be a Banach $\mathbb{Q}_p$-algebra. Then $A^{\leq 1} = A^\circ$, which we recall is the collection of power-bounded elements.

Example 56. Take $A \coloneqq \mathbb{A}_p\langle \mathbb{N}[S]\rangle$, which is the Tate algebra with S as variables. Then $A^{\leq 1} = \mathbb{Z}_p\langle \mathbb{N}[S]\rangle$.

Proposition 57. For nonnegative r and s, addition defines a map $A^{\leq r} \times A^{\leq s} \to A^{\leq \max\{r,s\}}$, and multiplication defines a map $A^{\leq rs}$.

Proof. Omitted. ■

Corollary 58. Let $A \in \mathrm{Ring}_{\mathbb{Q}_p}$ be bounded. Then $\mathrm{Nil}^\dagger A$ is an ideal of A.

Proof. Setting $r = s = 0$ in Proposition 57 quickly completes the proof. To be technical, we should check that $A = \bigcup_r A^{\leq r}$. To show this, note that $A^\circ \subseteq A^{\leq 1}$ by definition of $A^{\leq 1}$. But $A^{\leq 1} \subseteq p^{-1}A^\circ$ because $A^{\leq 1}$ is an intersection of all disks of radius greater than one. It follows that A, which is $A^\circ[1/p]$ can be written as $\bigcup_r A^{\leq r}$ by moving the powers of p around. ■

Definition 59. Fix a bounded $A \in \mathrm{Ring}_{\mathbb{Q}_p}$. Then A is †-*reduced* if and only if $\mathrm{Nil}^\dagger(A) = 0$. In general, we define the †-*reduction* as $A^{\dagger\text{-red}} := A/\mathrm{Nil}^\dagger A$.

Remark 60. It turns out that

$$\begin{array}{ccc} A^{\leq r} & \longrightarrow & A^{\dagger\text{-red},\leq r} \\ \downarrow & & \downarrow \\ A & \longrightarrow & A^{\dagger\text{-red},\leq r} \end{array}$$

is a pushout square, so $A^{\dagger\text{-red}}$ is bounded and †-reduced. However, the corresponding statement of A° is false in general. (In particular, this is a feature of Banach $\mathbb{Q}_p$-algebras.)

3.2 Uniform Rings

Intuitively, the uniform ring is analogous to taking a p-adic completion.

Definition 61. Fix a bounded $A \in \mathrm{Ring}_{\mathbb{Q}_p}$. Then define the *uniform ring* to be

$$A^u := \left(\lim_{r>0} A^{\leq 1}/A^{\leq r}\right)\left[\frac{1}{p}\right].$$

We say that A is *uniform* if and only if the natural map $A^{\leq 1}[1/p] \to A^u$ is an isomorphism.

Remark 62. Intuitively, taking the inverse limit is a p-adic completion. Roughly speaking, it will turn out to be the completion of $A^{\dagger\text{-red}}$.

Example 63. Fix a Banach $\mathbb{Q}_p$-algebra A. Then $A^{\leq 1} = A^\circ$, so A is uniform if and only if A° is p-adically complete. Here is a non-example: note $A := \mathbb{Q}_p[t]/(t^2)$ has $A^\circ = \mathbb{Z}_p \oplus \mathbb{Q}_p t$, which is not p-adically complete because $\mathbb{Q}_p t$ dies upon taking p-adic completion.

Example 64. Take $A := \mathbb{Q}_p\langle \mathbb{N}[S]\rangle_{\leq 1}$. Note that there is an inclusion $A \subseteq \mathbb{Q}_p\langle \mathbb{N}[S]\rangle$: on a test algebra $B \in \mathrm{Ring}_{\mathbb{Q}_p}$, this is given by the inclusions $B^\circ \hookrightarrow B^{\leq 1}$. This yields an inclusion $A^{\leq 1} \hookrightarrow \mathbb{Z}_p\langle \mathbb{N}[S]\rangle$, and it turns out that

$$A^{\leq 1} = A \cap \mathbb{Z}_p\langle \mathbb{N}[S]\rangle.$$

Thus, $A^{\leq 1}/p^n = \mathbb{Z}/p^n[\mathbb{N}[S]]$, so taking the uniform completion yields $A^u = \mathbb{Q}_p\langle \mathbb{N}[S]\rangle$.

Here are some basic properties.

Remark 65. It turns out that $A^u \cong \left(A^{\dagger\text{-red}}\right)^u$. Basically, this should follow from the definition and Remark 60, basically pulling out the reduction.

Proposition 66. Fix bounded $A \in \mathrm{Ring}_{\mathbb{Q}_p}$.

(a) The following is a pushout square in $D(\mathbb{Z}_p)$ for any $r \geq 0$.

$$\begin{array}{ccc} A^{\leq r} & \longrightarrow & A^{u,\leq r} \\ \downarrow & & \downarrow \\ A & \longrightarrow & A^u \end{array}$$

(b) One has $A^{u,\leq 1} = \lim_{r>0} A^{\leq 1}/A^{\leq r}$, so $(A^u)^u = A^u$.

(c) The ring A^u is $\dagger$-reduced.

(d) Choose $f \in A$. Then f is invertible if and only if its image in A^u is invertible.

Remark 67. By setting $r = 0$ in (a), we see that $A^{\dagger\text{-red}} \to A^u$ is injection.

Remark 68. The fact that $(A^u)^u = A^u$ means that $(-)^u$ is left adjoint to the inclusion $\mathrm{Ring}^u_{\mathbb{Q}_p} \subseteq \mathrm{Ring}^b_{\mathbb{Q}_p}$.

3.3 Gelfand Rings

We are now allowed to make a definition.

Definition 69 (Gelfand)**.** A bounded ring $A \in \mathrm{Ring}_{\mathbb{Q}_p}$ is *Gelfand* if and only if $A/A^{\leq 1}$ is discrete. Let $\mathrm{GelfRing}_{\mathbb{Q}_p}$ be the subcategory of Gelfand rings.

Remark 70. Being discrete is closed under filtered colimits, subquotients, and extensions. Thus, a ring A is Gelfand if and only if any (equivalently, all) of $A/A^{\leq r}$, $A/A^{<r}$, $A^{\leq r}/A^{\leq s}$ for $r > s$, etc. is discrete.

Example 71. If A is a Banach $\mathbb{Q}_p$-algebra, then $A^{\leq 1} = A^\circ$, which is open, so $A/A^{\leq 1}$ is discrete, so A is Gelfand.

Example 72. Consider $A := \mathbb{Q}_p\langle \mathbb{N}[S]\rangle_{\leq 1}$. Then A is Gelfand if and only if $A^{\leq 1}/p$ is discrete, which is $\mathbb{F}_p[\mathbb{N}[S]]$. But

$$\mathbb{F}_p[\mathbb{N}[S]] = \bigoplus_n \mathbb{F}_p[S^n/\Sigma_n].$$

If S is infinite, then the terms have infinite products of $\mathbb{F}_p$s, which fails to be discrete, so A fails to be Gelfand. But if S is finite, then the terms are finite products of $\mathbb{F}_p$, which is discrete.

Here is a better way to test to be a Gelfand ring.

Lemma 73. Fix a bounded ring $A \in \mathrm{Ring}_{\mathbb{Q}_p}$. Then A is Gelfand if and only if A^u is a Banach $\mathbb{Q} - p$-algebra.

Proof. By Proposition 66, we see $A/A^{\leq 1} = A^u/A^{u,\leq 1}$. Thus, we may assume that A is uniform. Now, A is a Banach $\mathbb{Q}_p$-algebra if and only if $A^{\leq 1}/p$ is discrete. For example, the reverse direction follows because one basically tries to reconstruct A from $A^{\leq 1}/p$ by taking some inverse limits, which means that A is a purely classical object and hence a Banach $\mathbb{Q}_p$-algebra. ■

Example 74. Recall $(\mathbb{Q}_p\langle t\rangle_{\leq 1})^u = \mathbb{Q}_p\langle t\rangle$. Thus, $\mathbb{Q}_p\langle t\rangle_{\leq 1}$ is Gelfand.

Remark 75. One can use this to check that $\mathrm{GelfRing}_{\mathbb{Q}_p}$ is closed under colimits. Because $(-)^u$ is left adjoint to the forgetful functor, we find that $(\operatorname{colim} A_i)^u$ is isomorphic to $(\operatorname{colim} A_i^u)^u$. The latter is now a colimit of Banach $\mathbb{Q}_p$-algebras and hence a Banach $\mathbb{Q}_p$-algebra.

3.4 The Berkovich Spectrum

Here is our construction.

Definition 76 (Berkovich spectrum)**.** Fix a Banach ring $(B, |\cdot|)$. Then the *Berkovich spectrum* $\mathcal{M}(B)$ is the set of multiplicative seminorms $\|\cdot\| : B \to \mathbb{R}_{\geq 0}$ dominated by $|\cdot|$. Then $\mathcal{M}(B) \subseteq \prod_{b\in B}[0, |b|]$, and we give $\mathcal{M}(B)$ the subspace topology. For a Gelfand ring A, we define $\mathcal{M}(A) := \mathcal{M}(A^u)$.

Remark 77. One can cut out $\mathcal{M}(B)$ from $\prod_{b\in B}[0, |b|]$ by some non-strict inequalities, so it is a compact Hausdorff space.

Example 78. Take $A = \mathbb{C}_p\langle t\rangle$. Then $\mathcal{M}(A)$ is a cofiltered limit of finite trees.

- Type I points: for each $x \in \mathcal{O}_{\mathbb{C}_p}$, there is a seminorm $f \mapsto |f(x)|$.
- Type II points: given $x \in \mathcal{O}_{\mathbb{C}_p}$ and $r \in [0, 1]\cap p^{\mathbb{Q}}$, there is a seminorm $f \mapsto \sup\{|f(x')| : |x' - x| \leq r\}$. Note that one gets the same seminorm if x is moved to any other point in the disk $\overline{B(x, r)}$.
- Type III points: given $x \in \mathcal{O}_{\mathbb{C}_p}$ and $r \in [0, 1] \setminus p^{\mathbb{Q}}$, one can write down the same formula. These points connect type II points (and type I points).
- Type IV points: given a decreasing sequence of nonempty disks $\{D_n\}$ for which $\bigcap_n D_n = \varnothing$, one can define the seminorm $f \mapsto \inf_n \sup_{x\in D_n} |f(x)|$.

Type I and type IV points are "leaves" of the tree. Note that in type IV, the radii of the disks must tend to a positive number, so they are arguably not at the "bottom."

4 October 26: Gelfand Rings and Gelfand Stacks

This talk was given by Daniel and then Jeremy at MIT for the Analytic ring stacks seminar.

4.1 Fredholm Rings

Fix a presentable symmetric monoidal stable ∞-category $\mathcal{C}$.

Definition 79. Fix a symmetric monoidal stable category $\mathcal{C}$. Then the subcategory of *perfect objects* is the stable subcategory generated by 1, finite limits, shifts, and retracts.

Remark 80. Note that this subcategory is also closed under finite colimits by stability.

Definition 81. Fix a symmetric monoidal stable category $\mathcal{C}$. Then the subcategory of *dualizable objects* consists of those $c \in \mathcal{C}$ admitting $c^\vee \in \mathcal{C}$ and evaluation $\mathrm{ev}\colon c \otimes c^\vee \to 1$ and co-evaluation maps $\mathrm{coev}\colon c^\vee \otimes c \to 1$ satisfying some conditions. Namely,

$$c = 1 \otimes c \to (c \otimes c^\vee) \otimes c = c \otimes (c^\vee \otimes c) \to c \otimes 1 = c$$

is the identity, and similarly for $c^\vee$.

Example 82. The unit is self-dualizable.

Remark 83. It turns out that perfect objects are always dualizable. In $D(R)$ where R is a classical ring, the converse holds.

Today, we are working with some general analytic rings $A = (A^\blacktriangleleft, D(A))$.

Definition 84. An analytic ring A is *Fredholm* if and only if dualizable objects are perfect in $D(A)$.

Remark 85. Recall that $A^\blacktriangleleft \in D(A)$, so the perfect objects in $D(A^\blacktriangleleft)$ live in $D(A)$. Thus, being Fredholm is equivalent to saying that dualizable objects in $D(A)$ all come from the smaller algebraic category $D(A^\blacktriangleleft(\mathrm{pt}))$.

Example 86. If A comes from a complete Huber pair, then A is Fredholm.

Example 87. If A is a Gelfand ring, then A is Fredholm.

Non-Example 88. The condensed ring $A = \mathbb{Q}_p[t]$ fails to be Fredholm. Roughly speaking, the problem is that A is too big. Define $P \coloneqq \left(\prod_{\mathbb{N}} \mathbb{Z}_p\right)[1/p]$, which lives in $D(\mathbb{Q}_p)$. We now claim that

$$M \coloneqq \mathrm{cofiber}\left(1 - t\left(1, p, p^2, \ldots\right); P \otimes_{\mathbb{Q}_p} \mathbb{Q}_p[t]\right)$$

exhibits our counterexample. The moral is that the endomorphism $\left(1, p, p^2, \ldots\right)$ is trace class. One can use some functional analytic arguments to show that M is dualizable. However, M has nontrivial pullback to the fiber $t = p^{-n}$ for all $n \in \mathbb{N}$, so it has fibers going off to infinity, so it fails to be perfect.

4.2 More on the Berkovich Spectrum

Last time, we introduced the Berkovish spectrum of a Gelfand ring A.

Definition 89. Fix a Gelfand ring A. Given elements $f_1, \ldots, f_n, g \in A(\mathrm{pt})$ which generate $A(\mathrm{pt})$ as an ideal (in π_0). Then we define the associated closed rational subspace

$$\mathcal{M}\left(\frac{f_1, \ldots, f_n}{g}\right) \coloneqq \left\{|\cdot| \in \mathcal{M}(A^u) : |f_i| \leq |g| \neq 0 \text{ for each } i\right\}.$$

Proposition 90. Fix a Gelfand ring A. Suppose $Z \subseteq \mathcal{M}(A)$ is rational closed. Then there is a Gelfand idempotent A-algebra A_Z such that any map $A \to B$ of Gelfand rings extends to $A_Z \to B$ if and only if the associated topological map $\mathcal{M}(B) \to \mathcal{M}(A)$ land in Z.

Proof. Write $Z = \mathcal{M}(f_1, \dots, f_n/g)$. Then define $A_Z \coloneqq A\langle t_1, \dots, t_n\rangle_{\leq 1}/(gt_i - f_i)_i$. ■

Remark 91. One can check that $\mathcal{M}(A_Z) = Z$.

There is a more categorical way to work with such spaces.

Definition 92 (smashing spectrum)**.** Fix a stable category $\mathcal{C}$. Then the *smashing spectrum* $\operatorname{sm}\mathcal{C}$ is the collection of idempotent algebras in $\mathcal{C}$.

Remark 93. This is a locale, which is some kind of weakening of a Grothendieck topology.

Example 94. If A is a Noetherian ring, then $\operatorname{sm} D(R)$ is isomorphic to $\operatorname{Spec}(R)^{\mathrm{op}}$ as locales. However, this is rarely true outside the Noetherian setting.

Remark 95. Fix a Gelfand ring A. Then there is a unique natural map of locales $\operatorname{Sm} D(A) \to \mathcal{M}(A)$ extending $Z \mapsto A_Z$. Here, being a map of locales means that closed subsets have pre-image given by closed sets.

Notation 96. Fix a profinite set S. Then we define S_{Betti} to be the animated spectrum of $C_{\mathrm{cts}}(S, \mathbb{Z})$.

Remark 97. By taking colimits, this construction extends to a functor $(-)_{\mathrm{Betti}}$ from condensed animated modules to animated stacks.

Remark 98. Fix a Gelfand ring A. When $\mathcal{M}(A)$ is metrizable and finite-dimensional, then there is a natural map

$$\operatorname{AnSpec} A \to \mathcal{M}(A)_{\mathrm{Betti}} \times_{\operatorname{AnSpec}\mathbb{Z}} \operatorname{AnSpec}\mathbb{Q}_p$$

which satisfies universal !-descent. The hypotheses on $\mathcal{M}(A)$ are needed to be able to map into it (as some kind of compact Hausdorff space). ■

4.3 Gelfand Stacks

We now switch to Jeremy.

Definition 99 (separable)**.** A Gelfand ring R is *separable* if and only if R^u has a countable dense subset.

Remark 100. This is equivalent to saying that R^u is ω_1-compact in Banach $\mathbb{Q}_p$-algebras. Roughly speaking, separability is a finiteness property.

Definition 101 (Gelfand stack)**.** Define the category of *Gelfand stacks* to be the category of analytic stacks on the opposite category of Gelfand rings.

Notably, the finiteness condition lives on the uniform completion, which intuitively erases any (topologically) nilpotent fuzz.

These Gelfand stacks are supposed to contrast with arc stacks.

Definition 102 (arc cover)**.** A map $A \to B$ of perfectoid Tate rings over $\mathbb{Q}_p$ is an *arc cover* if and only if $\mathcal{M}(B) \to \mathcal{M}(A)$ is surjective.

Definition 103 (separable)**.** A Tate ring A is *separable* if and only if A°/ϖ is countable, where $\varpi \in A$ is a uniformizer.

Remark 104. Again, an affine perfectoid $\mathbb{Q}_p$-algebra A is separable if and only if it is ω_1-compact.

Definition 105. The category of *arc stacks* is the category of hypersheaves on the arc site of the category of $\mathrm{AffPerfd}_{\omega_1}^{\mathrm{op}}$.

Remark 106. Each perfectoid Tate algebra A admits an arc cover $A \to B$ where B is totally disconnected (i.e., $\mathcal{M}(B)$ is profinite).

Here is our main construction, which we will return to next week.

Proposition 107. Fix a separable Gelfand ring A. Define the functor $\mathrm{Mor}_{\mathrm{GelfRing}}(A, -)$, restricted to totally disconnected objects in affinoid perfectoid ω_1-compact rings. Then this functor is an arc stack.

Taking descent produces the following.

Corollary 108. There is a functor $(-)^\diamondsuit : \mathrm{GelfStk}_{\mathbb{Q}_p} \to \mathrm{ArcStk}_{\mathbb{Q}_p}$.

This functor is supposed to be destructive, getting rid of nilpotence in some sense. Next time, we will show that $(-)^\diamondsuit$ has a right adjoint $(-)^{\mathrm{DR}}$.